

Recent advances in Effective Remediation of Groundwater Fluoride

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Abstract: Fluoride is generally referred to as a double-edged sword as dental caries are related to dental caries, but enormous intake causes dental and skeletal fluorosis. Fluoride in ground water comes due to dissolution of natural minerals in the rocks, soils into which water interacts. Toxic fluoride concentration in drinking water is noted in several areas around the world and millions of people depend on groundwater with range of concentrations exceeding the World Health Organization (WHO) guideline level of 1.5 mg/L. In India, 65 million individuals are believed to be affected by endemic fluorosis. With a proposal to resolve the problem, the Rajiv Gandhi National Drinking Water Mission has been established to monitor fluorosis. Nalgonda technique, Prashanti activated alumina technology, UNICEF in India using defluoridation based on households have been developed. A simple and economical domestic defluoridation process was developed by various researchers on the basis of findings and comprehensive investigations. This paper seeks with recent techniques on defluoridation studies using traditional and unconventional materials and to compile the various pros and cons of these defluoridation techniques, including prashanti activated alumina technology, membrane method, water treatment residues (WTR), $\text{Fe}_3\text{O}_4/\text{Al}_2\text{O}_3$ nanoparticles adsorption using red mud, clay, bauxite, clay, *Mentha longifolia*.

Key Words: Emerging trends, Defluoridation, Ground water, Fluorosis, India

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2.1. Introduction

Entry to healthy drinking water is a fundamental human right and one of the most important indicators of development. Groundwater is the most valuable natural resource and national asset, and it has taken a predominant role in the overall water resource. Groundwater is significant and invariable sources of drinking water around the globe [1]. It is estimated that 1.5 billion people across the world rely on groundwater for drinking and irrigation. Groundwater provides about 80% of all agricultural water used for domestic purposes in India, with about 39 million hectares irrigated by 45 percent of all agricultural water. Groundwater quality data demonstrates rock geological history, groundwater recharge, discharge, climate, movement, storage. Groundwater is India's main water supply for domestic, irrigation, and industrial uses. Groundwater is an effective tool for determining pollution levels and plays a pivotal role in many aspects of the food chain [2-4].

Due to a shortage of surface water in India, most of people rely on groundwater for a variety of purposes, including drinking, domestic, irrigation, and industrial. Due to this, the availability of safe drinking water is essential for human development [5]. Groundwater is the ultimate natural resource and a vital national commodity, and it has taken on a significant role in the overall water resource. However, rapid urbanization and industrialization have resulted in environmental pollution and health effects due to increasing population, new technical advances, and the depletion of groundwater resources. Owing to a lack of effective waste management methods, many anthropogenic activities and their byproducts have contaminated surface and subsurface water. Water quality is influenced by effluents from industries, agriculture, and horticulture transportation. Hence, the quality of drinking water is critical in a variety of health concerns.

Contamination of Fluoride in ground water has been an important public health concern around the world for the past two decades. Because of their importance, such work has been taken up all over the world in arid and semi-arid regions. Greater fluoride concentration in groundwater is one of the main health consequences in several countries. Fluoride levels are higher in greater than 23 countries around the world, especially in places like India, Eastern Africa, Central

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Africa, China, and South America [6-9]. Numerous groundwater quality related work involving fluoride have been conducted in India, especially in arid and semi-arid regions [10-12]. Groundwater is majorly used for drinking and agricultural practices in South India's semi-arid zone, and several quality studies have been conducted [13-27]. The concentration of fluoride is governed by a variety of factors, such as availability of fluoride-bearing rocks, rock composition, the mechanism of rock water interaction, evaporation, evapotranspiration, the appearance of aluminium fluoride ions, beryllium, ferric acid, and a series of mixed fluoride hydroxide cofactors [28]. Fluorides are found in three major forms: fluorspar or calcium fluoride (CaF_2), apatite or rock phosphate ($\text{Ca}_3\text{F}(\text{PO}_4)_3$), and cryolite (Na_3AlF_6)[10], with higher fluoride dosages (1.5 mg/L) and fluoride concentrations (>3.0 mg/L) required for normal bone mineralization and dental enamel formation [13].

Fluorine is the lightest of the halogens and the 13th most abundant naturally occurring element in the Earth's crust. Since it is the crucial electronegative and reactive of all other elements, Elemental fluorine is only present as fluoride mineral complexes in nature. Fluorides make up 0.06–0.08% of the Earth's surface, but they have a low average abundance (300 mg kg⁻¹); [6]. Moreover, unlike any of the other halogens, the bulk of fluoride concentration in the Earth's crust comes from lithology, with just a limited amount originating from other sources like air, seawater, and other anthropogenic practices [10]. Fluorine is both the most electronegative and reactive of all known elements: electronegativity 4.0; It has a 626 g/g abundance in the continental crust [21]. Fluorine is rarely occur as free in nature, but it is commonly found as the fluoride ion in minerals (F^-). The presence of fluorine in groundwater has sparked global interest owing to its major effect on human physiological conditions [14]. Fluorine is found in natural waters primarily as the free fluoride ion, while fluoride complexes of Al, Be, B, and Si can also be found under certain prevailing conditions. Fluoride ingestion into the human body from potable water at a concentration of 1 mg/l facilitates bone growth and arrests dental caries. The World Health Organization [5] framed a maximum tolerance limit for fluoride level in drinking water at 1.5 mg/l.

2.1.1. Public Health Importance

Fluoride not only has remarkable chemical properties, and also has physiological properties that are important to human and animal health. Fluoride ions can inhibit or activate enzymatic processes even at very low concentrations. Phosphatase, lipase, and urease enzymes are all inhibited by it. Fluoride has been shown to interact with thyroid function (Teotia et al, 1972). On the other hand, it is found to be useful in the natural phase of mineralisation of hard tissues, such as bones, also in trace quantities. Fluoride has been shown to interact with thyroid function (Teotia et al, 1972). On the other hand, it is found to be useful in the natural phase of mineralisation of hard tissues, such as bones, also in trace quantities. Fluoride in drinking water up to 1.0 mg/l is advantage to children's health because it facilitates the production of caries-resistant dental enamel. Fluoride in drinking water above 1.0 mg/l has been shown to be detrimental to human and animal physiology, aside from a few beneficial effects. The toxic effects on teeth include chalky white patches, pitting, and chipped off tips, as well as premature tooth loss (Fig:2.1). Skeletal defects and joint pain occur at concentrations greater than 2-3 mg/l. Ingestion of the ion for an extended period of time causes spinal weakness and kyphosis (Stooping forward). Symptoms of dental and bone fluorosis include being unable to squat and being bedridden for the remainder of one's life (Jolly et al, 1970). In addition to humans, cattle are affected by fluoride ion ingestion. The seriousness of the fluorosis issue is exacerbated by the fact that skeletal changes are almost often permanent and the condition is incurable.



Figure 2.1: Dental Fluorosis

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2.2. Methods

2.2.1. Fluoride Ion Selective Electrode

The use of an ion selective electrode for fluorides is the noted popular electro analytical approach in determining fluoride ions in water solutions. The fluoride selective electrode is extremely sensitive, with a temperature range of 0 to 50°C. In commercial use, electrodes with standard homogenic membranes made from fluoride lanthana (LaF_3), which were first denoted by Frant and Ross, are commonly applied for potentiometric analysis of fluoride ions (Frant and Ross, 1966; Frant, 1994).

2.2.2. Theoretical structure

The potentiometric approach is focused on possible measurements to determine a galvanic element's electromotive power. Ion selective electrodes (ISEs), that are capable of rapid and selective measurements of analyte levels are almost exclusively used in direct potentiometric determinations.

2.2.3 Materials and Processes

NaF dried for 1 hour at 100°C. The composition of liquid NaF is uncertain. Fluoride ISE with KCl (7.55 g) and Ag/AgCl or SCE for reference electrode. A multimeter or pH meter is required that can show mV potentials is needed (Figure. 2.2).



Figure. 2.2: Ion selective electrode (Orion 4 star ion meter, Model: pH/ISE).

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The fluoride electrode is used to perform a simple, inexpensive, and accurate potentiometric determination of fluoride content in solutions. The use of TISAB buffer eliminates any negative interfering effects that could affect the process of determination. The fluoride content of samples is measured using direct potentiometry and, in the case of extremely low values, the normal addition process. When it came to determining fluoride ions in water, Method ISP proved to be more efficient, fast, and reliable. It also has an advantage over other analytical methods in that it is non-destructive and allows samples to be used for other purposes. The most popular and convenient methods of fluoride and chloride ion determination tend to be electro analytical methods based on potentiometry with ion-selective electrodes. Because of their high selectivity, precision, and low detection limits, fluoride selective electrodes can be used to assess fluoride concentrations in drinking water. The study's benefits include a fast analysis time, the elimination of sample pretreatment, the measurement system's simplicity, and a relatively low instrument cost.

2.3. Technologies of Defluoridation

To comprehend the environmental effects of altering fluorine mechanisms, there is an emerging trend to first comprehend fluorine's natural cycling. This section summarizes about this element, its environmental mechanisms, actions of its diverse types. Although this topic seems to be unrelated to fluoridating a public water source, it will provide insight into the chemistry and reactivity of fluorine in natural environments, as well as possible reactions with pipes and additives in the water supply. It also considers what happens to the element after it is released into the environment through wastewater treatment plant effluent. Fluorine (F^-) is a halogen with an atomic mass of 18.998, an atomic number of 9, and a valence of one. In its pure form, it is a diatomic gas (F_2). Fluorine is the most electronegative element known, and it is a very strong oxidizer in its gaseous form, F_2 . Fluorine occurs in nature as fluoride ion due to its strong oxidizing ability (F^-). The most popular and environmentally reactive ionic form is this one. Fluorine's natural concentration in the earth's crust varies between 0.06 to 0.09 percent by weight. Despite the fact that fluorine is a necessary element, Fluorine is an important ingredient for mammals, but at high concentrations, it is poisonous to both animals and plants. The purpose of fluoridation is to add fluorine in its ionic (fluoride) form to water sources. To accomplish this, a variety of compounds are used, all of which dissociate into their respective ionic forms and

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enter equilibrium with the chemistry of the receiving water. Fluoride flows at a rate of 3.7 million metric tons per year from continental drainages to the oceans in the environment. Mining extracts 2 million tons per year, while combustion of fossil fuels and industrial pollution release 0.022 million tons (as HF, SiF₄, and metal fluorides) into the atmosphere. Dust blown by the wind and volcanic emissions are also sources of pollution in the atmosphere (Gaciri and Daries, 1993). Fluorine precipitates in marine sediments that could be exposed over geologic time, allowing fluorine to be returned to continental landmasses. Fluorine from the atmosphere is mainly recycled to the earth through precipitation.

The aim of fluoride removal was to treat contaminated water and lower the fluoride content to safe levels. Defluoridation is a notable process of removing fluoride ions from potable water. The various methods of extracting excess fluoride from water that have been attempted can be divided into four categories as follows: a) methods of adsorption, b) ion exchange mechanisms, c) methods of precipitation, and d) miscellaneous methods. Some defluoridation techniques were developed to arrest fluoride content in water using adsorbents such as sunflower plant dry powder, phytomass steam, neem bark powder, activated carbon, cotton jute, bagasse ash, burnt bone powder, phosphate treated saw dust, bone char, and others. Defluoridation was also achieved using the Nalgonda procedure, process of activated alumina, and the ion exchange process. Moreover these techniques are not commonly used due to their high cost, low performance, or inapplicability on a large scale. Because of the ever-increasing population and the increased demand for safe drinking water, the advent of treatment of harmful components, particularly fluoride in fluid solution, has become vital source of concern for them since then. Fluoride exposure originated by natural, anthropogenic practices the fluid framework, also it causes a number of diseases such as joint pain, malignant development, weak bones, and damage to the cerebrum.

The current research focuses on various fluoride removal mechanisms, viz., adsorption, ion exchange, membrane filtration, and electrocoagulation, that are used to remove fluoride from fluid media. However, using various techniques like ion exchange, layer filtration, and electrocoagulation, evacuation productivity was increased to 90–95 percent, 99 percent, and 85.5 percent, respectively. Adsorption has been broadly investigated as an extremely viable and

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emergent technique because of easy operation, minimal effort, fair efficiency. However, using various techniques such as ion exchange, layer filtration, and electrocoagulation, evacuation productivity was increased to 90–95 percent, 99 percent, and 85.5 percent, respectively. The findings of this study clearly show that each process has discovered a new mechanism for the remediation of harmful fluoride particles [38].

Because of increased awareness and demand for potable water to address the problems of the developing population, especially in developing countries like India, fluoride remediation from drinking water has become an undeniably notable issue of open concern. This paper provides an overview of recent advancements in defluoridation techniques, as well as a discussion of the issues and research opportunities in the vital area of fluoride removal. X-Ray diffraction (XRD), field emission scanning electron microscopy (FESEM), energy-dispersive X-ray spectroscopy (EDX), transmission electron microscopy, Fourier transform infrared spectroscopy (FTIR), inductively coupled plasma mass spectrometry, atomic absorption spectrometry, and inductively coupled plasma-optical emission spectroscopy are the spectroscopic methods used in this study to produce adsorbents/adsorption devices [39].

2.3.1. Prasanti Technology Applying Activated Alumina

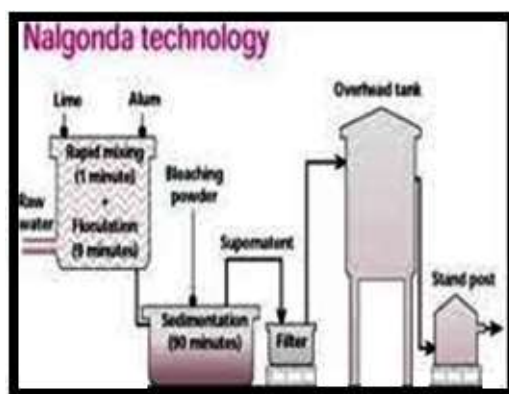


Figure 2.3: Nalgonda Technique

Prasanti's activated alumina fluoride removal technology is currently applied for water defluoridation in Indian villages [40]. It was established due to research and development

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activities at the Satya Sai Higher Learning University's Department of Bioscience at Prasanti Nilayam, Puttaparthi (Anantapur District) of Andhra Pradesh, South India (Figure 2.3).

2.3.2 Membrane technology

Despite the fact that many traditional water purification methods are currently being used to resolve the issue of groundwater pollution, none of them has been found to be a user-friendly or cost-effective solution due to some or other limitations, and none of them has a payback period of zero or very long.

2.3.3. The membrane mechanism of reverse osmosis (RO)

Over the last decade, the RO membrane system has developed itself as a notable method of supplying safe drinking water without the problems associated with other conventional methods [41]. RO is a physical process for removing contaminants by adding pressure to the feed water and guiding it with a semi-permeable membrane. The mechanism is the polar opposite of natural osmosis since pressure is generated to the concentrated side of the membrane, which overcomes the usual osmotic pressure.

Numerous studies [42-46] have demonstrated fluoride removal efficiencies of up to 98 percent using membrane processes. This property of nano-filtration in comparison to reverse diffusion can have certain advantages, and experimental and theoretical studies are devoted to a material retention mechanism to facilitate the production and selection of targeted membranes as well as condition optimization [47]. As the pH of the water is increased from 5.5-7, the fluoride removal in the reverse osmosis phase has been stated to range from 45-90 percent. The economics of a plan is often evaluated in particular circumstances. There are some disadvantages to this process, such as the extraction of all water ions, but it is very good for defluoridation. Although certain minerals are required for optimal growth after treatment, remineralization is needed.

2.3.4. Residues from the Treatment of Water (WTR)

WTR has been shown to be an adsorbent for the repair of many anions found in water and soil (perchlorate, selenium, and arsenic) and may be able to extract fluoride from polluted water. In

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order to meet the drinking water criteria for fluoride removal capacity, safe reuse of publicly accessible waste from drinking water treatment plants, namely WTR, has been used. Precision surface region, Fourier transform infrared (FT-IR), scanning electron microscopy, and X-ray powder diffraction have all been used to characterize WTRR (XRD). To determine the best adsorption strength, batch adsorption experiments were conducted using WTR dosage, contact time, agitation velocity, initial levels of fluoride concentration, temperature, specific area and water pH. The metal hydroxyl and metal oxide groups generated to the elimination of fluoride, according to the FT-IR and XRD experiments. The use of WTR will also help with defluoridation and provide a means of reuse for the treatment of drinking water waste [48].

2.3.5. $\text{Fe}_3\text{O}_4/\text{Al}_2\text{O}_3$ Nanoparticles:

Using Fe_3O_4 and Al_2O_3 NPs that were green synthesised with jojoba defatted meal extract as a reducing agent, a low-cost defluoridation technique was developed (Sonu kumara et al., 2017). These NPs were impregnated into polyurethane foam (PUF) and used to make tea infusion bags.

The Al_2O_3 NPs-PUF showed a higher water defluoridation capacity of 43.47 mg g^{-1} of F compared to 34.48 mg g^{-1} of F with Fe_3O_4 NPs-PUF. The F was 1.5 mg L^{-1} below the appropriate amount before the synthesised Al_2O_3 -PUF infusion bags were used. The F was 1.5 mg L^{-1} below the appropriate amount before the synthesised Al_2O_3 -PUF infusion bags were used. The effect of various parameters like pH, contact time, PUF scale, and initial F concentration has been verified through sorption experiments. A mixture of FESEM, EDX, XRD, and FTIR techniques is used to describe the adsorbent's different properties. The advantages of the established method described in this study are high F removal capability, comprehensibility, portability, sustainable, viable making this approach perspective for constrained environments, mainly in rural areas with high F groundwater pollution [49].

Different groups investigated the optimal adsorption potential of different materials under similar conditions. Under similar conditions, the adsorption potential of carbon nanotubes, typical soil, g- Al_2O_3 , and activated carbon follows the order $\text{CNT} > \text{soil} > \text{g-}\text{Al}_2\text{O}_3 > \text{activated carbon}$; while SWCNs, had a higher fluoride removal efficiency than MWCNs, which was also very high in optimal conditions with two forms of fine powder and 150 mesh activated alumina. Under

similar conditions, the adsorption potential of alumina-infused carbon nanotubes was observed to be 13.5 times that of AIC-300 (4 times) g-Al₂O₃ and IRA-410 polymeric resin. Under ideal conditions, the overall adsorption potential of charcoals was discovered to be bone char > coal charcoal > wood charcoal > carbon black > petroleum coke in the same order (Figure 2.4.).

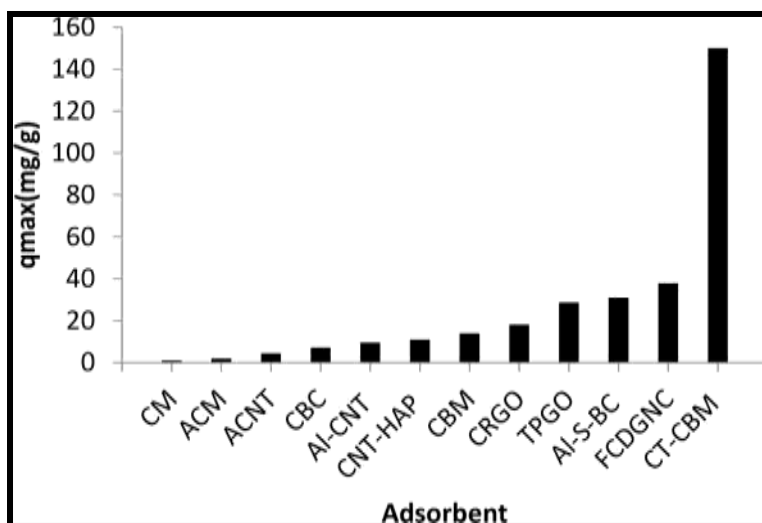


Figure 2.4. Comparison of carbon-based materials adsorption ability for water defluoridation.

2.3.6. Techniques for Adsorption

This technique is based on the adsorption of fluoride ions to the surface of an active agent. In the process of adsorption, raw water is allowed to pass through a series of bed containing defluoridating material. Physical, chemical, or ion exchange processes are used to keep fluoride in the content. After a certain period of action, the adsorbent becomes saturated and needs regeneration [50].

2.3.6.1. Activated alumina

Low-temperature dehydration (300-600°C) of aluminium hydroxides creates activated alumina (Al₂O₃), which has been applied for defluoridation since 1934. Fluoride is extracted by a ligand exchange reaction on the surface of activated alumina. Alumina (aluminum oxide, Al₂O₃) is nearly insoluble in water. The solubility is determined by previous heat treatment method. Various activated grades of alumina are available at a low cost in the United States. The porosity and surface area of the alumina determine the suitability of the defluoridation grade [51].

2.3.6.2. Alumina Plus Manganese Dioxide.

Manganese dioxide is a major insoluble sorbent that is non-hazardous. It is not used explicitly because, in comparison to the current work with dispersion over disposed earthenware, it may be needed in greater quantities (DEW). Alumina and manganese dioxide are combined. Manganese dioxide is an essential sorbent that is either insoluble or toxic. It is not allowed to be used because, in comparison to the current work with dispersion over disposed earthenware, it may be needed in larger quantities (DEW). Alumina and Manganese Dioxide combined. Manganese dioxide is an integral sorbent that is neither soluble nor toxic. It is not legal to use because, in comparison to current work with dispersion over disposed earthenware, it may be needed in larger quantities (DEW). Alumina with a MnO_2 coating [52-54] MnO_2 in bentonite clay [55] Enabled carbon with MnO_2 coating [56-57] Many studies have found KMnO_4 -coated activated carbon [58] and hydrated oxides of manganese ores to be relevant to fluoride removal. The removal of DEW fluoride at a pH of 7.15 with 0.025 percent MnO_2 was critical. Rinsing the spent sorbent with 1.5 M NaOH was found to be successful [59], suggesting that the reaction used to remove fluoride is reversible.

2.3.6.3. Bauxite

Widely available geomaterials which have been researched in recent years in order to determine an efficient solution for extracting high water fluoride content [60]. Bauxite is a readily available mineral that mainly contains alumina, iron, silica, and titanium oxides. As indicated by [54], several similar studies have revealed the removal of Cr(VI) and As(III). F was removed from aqueous solutions using thermally activated titania rich bauxite. Previous studies have shown that heat-activated bauxite has the potential to remove anionic pollutants including Cr(VI) and As(III)[61]. A bauxite sample from India's Eastern Ghats (Orissa) was used to defluoridate synthetic and fluoride-bearing groundwater obtained from fluoride-endemic villages near Tarabalu, Atri, and Taptapani hot springs of Orissa [62-63]. As bauxite is a prevalent mineral in several parts of the world, defluoridation can be a quick, reliable, and low-cost process. Bauxite is a strong material with a lot of mechanical power, and it's abundant, so it's a good choice for fluoride removal over other adsorbents. Finally, fluoride sorption processes on bauxite are a composite process that involves intra-particle diffusion, physisorption, and chemisorption [64]. The high alumina content bauxite tested for this study denoted a promising property as a low-cost and active adsorbent for the

Defluoridation. According to the findings, bauxite is a readily accessible material that can be used without the need for pre-activation, which is needed for alumina and other adsorbents. Based on factors like the active phase of modified bauxite, treatment system, pH balance, and initial concentration, several processes have been implicated in the process of fluoride sorption on bauxite, including ligand exchange, ion exchange, intra-particle diffusion, cation exchange and surface complexation [65].

2.3.6.4. Red mud (RM)

The solid waste residue produced during the caustic digestion of bauxite ores during the manufacture of alumina is known as red mud (RM). With a pH of 10-12.5, red mud is a highly soluble waste material. Tiny fragments of aluminum, iron, silicon, titanium oxides, and hydroxides make up the majority of red mud. It has a profound effect on waste disposal owing to its alkaline nature, chemical and mineralogical species, which poses a major challenge when alumina industries are developed. The application of red mud in the treatment of soil polluted by groundwater would be extremely beneficial [63]. It's also worth noting that red mud contains a lot of metal particles and organic substances that are released under unusual circumstances. However, it is important to note that red mud contains some metal ions as well as organic matter, all of which can be released under some conditions.

2.3.6.5. Clay minerals

Clay minerals are the chief natural adsorbents due to their small particle sizes, lamellar structures, and negatively charged surfaces. They are strong cation adsorbents with a broad specific surface region because of these characteristics. Adsorbents made of fired clays are inexpensive to use since they are easy to prepare and regenerate (firing temperature below 1000 °C). Clay and several clay minerals can thus be useful adsorbents in the purification of the environment [66-67]. The fluoride binding effect of basic pottery has been studied by a number of researchers. Fluoride binding capacity in 80 mg/kg clayware has been documented, and results have demonstrated that fluoride absorption in clayware is slow and finite [66]. Mud pots' key benefits are their cost-effectiveness and their ability to be affordable in rural areas [67].

2.3.7. Sorption study

Adsorption experiments are used to obtain rate and equilibrium data using the batch technique. The tests are performed by mixing a 10 g/L adsorbent dosage of 50 mL of aqueous solution with a known fluoride ion concentration and shaking the samples at a rate of 200 strokes/min with a shaking tool. By adding 0.5N HNO₃/0.1M NaOH to fluoride-containing samples, the pH is maintained at the optimal level (5-6). All of the measurements are performed at room temperature (29.5 ° C). The absorption of fluoride ions by the adsorbent material at their maximum pH values appears to be very rapid, according to preliminary investigations. Within the first hour of interaction with fluoride ions at a concentration of 10 mg/L and an adsorbent dose of 10 mg/L for treated bio-sorbents, 80 percent of the adsorption usually occurs [16].

2.3.7.1. Mechanism of Sorption

The Freundlich and Langmuir models are associated with fluoride ion removal sorption data [68]. Experiments on the adsorption capacity of *Mentha longifolia* (Mint) from aqueous solutions were studied at room temperature. The effects of many major parameters on adsorbent materials, like pH, contact duration, adsorbent quantity and particle size, and concentration of absorption fluoride ions, was investigated from a kinetic standpoint. The rate and equilibrium results are obtained using a batch technique in adsorption experiments. The sorption properties were calculated using the Freundlich and Langmuir isotherms. *Mentha longifolia* (Mint) has been shown to be efficient in extracting fluoride from groundwater samples [16].

2.4. Conclusion

Fluoride-polluted ground water is causing health issues in India. In this paper, we discussed different strategies for addressing the issue of higher fluoride levels in drinking water, including RO, nanofiltration, biopolymers, and others. Application of membrane technique is a low-cost and straightforward method for removing fluoride. Enabled alumina removal of fluoride is a tried-and-true treatment method. Because of the solubility of aluminum, immoderate application of aluminum salts generates sludge and has serious health consequences. The membrane technique is a low-cost and straightforward method for removing fluoride. Fluoride removal by activated alumina is a tried-and-true treatment method. Ion Exchange extracts up to 90% of fluoride thus leaving the water's taste and color unchanged. However, higher costs are a

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drawback, and treated water with a low pH and greater chloride content. The WHO and the USEPA guidelines enabled alumina adsorption to be one of the most effective fluoride removal technologies available. Attempts have been made as in the past to make modified alumina by infusing with metal oxides that have proven to be of significant defluoridation capacity. However, the ability to completely remove carbon-based fluoride technologies is still limited, and numerous experiments have been performed to alter the surface of defluoridation-enhancing carbon-based adsorbents. Moreover, some drawbacks have been recorded, including difficulties with regeneration and poor performance at high fluoride concentrations. Defluoridation was shown to be an inexpensive defluoridation technique when applied with Fe_3O_4 and Al_2O_3 NPs synthesized green using jojoba defatted meal extract as a reduction agent. The benefits of the proven technique described in this study include higher F removal capability, sensibility, portability, eco-friendly, viable, and hence making this strategy more fetching for resource-limited environments, particularly in rural areas where high groundwater fluoride pollution causes health issues. Biosorbents, especially chitosan, provide satisfactory results in terms of fluoride removal after modifications. Locally available adsorbent materials with good performance with cost effectiveness can be used for adsorption. This technique makes use of herbal and indigenous items that are both cost-effective and readily available. These products are acceptable for use in Indian conditions. With the presence of sulphate, phosphate, and bicarbonate, the process is pH-dependent and leads to ionic rivalry. The disposal of fluoride-laden sludge is a downside. Membrane processes are extremely effective and do not require the use of chemicals. It can function over a large pH spectrum and has little interaction with other ions. It does, however, necessitate higher costs and skilled labor. For water with high salinity, this method might not be appropriate. The ability to extract fluoride varies depending on a number of chemicals, geographical, and economic conditions unique to the region, resulting in actual installations that vary from generalizations. Any approaches that are effective in one field cannot be effective in other areas. As a result, all technology that will be used should be treated well before being deployed in the field.

2.5. References:

- [1]. P. Li, H. Qian, (2018) Water resources research to support a sustainable China, ***International Journal of water Resource Development***, 34(3): 327-336.
- [2]. V. Sunitha, Y. Sudharshan Reddy, (2019) Hydrogeochemical evaluation of groundwater in and around Lakkireddipalli and Ramapuram, Y.S.R District, Andhra Pradesh, India, ***Hydro Research***, <https://doi.org/10.1016/j.hydres.2019.11.008>.
- [3]. A. Narasimha, V. Sudharshan (2017) Assessment of fluoride contamination in groundwater from Basara, Adilabad District, Telangana State, India, ***Applied Water Science*** 7(6):2717-25,
- [4]. B. Suvarna, Y. Sudharsan Reddy, V. Sunitha, M. Prasad, (2018) Water quality Index of Ground water in and around Lakkireddipalli and Ramapuram, Y.S.R District, A.P India, ***Journal of Emerging Technologies and Innovative Research***, 5 (11): 786-794.
- [5]. World Health Organization (WHO), (2011) Guidelines for drinking-water quality, WHO chronicle, 38(4): 104-8.
- [6]. A.K. Susheela, (1993) Prevention and control of fluorosis in India, Rajiv Gandhi National Drinking Water Mission, Ministry of Rural Development, New Delhi, ***Health Aspect***, 1:20-22.
- [7]. Shahriari T et al, (2010) Evaluation of fluorine concentration in drinking-water sources in South Khorasan 2008–2009, ***Journal of Birjand University Medical Sciences***, 17:33–41.
- [8]. A. Jafari, M. Amir Hossein, G. Hatam, R. Reza, S.H. Sara, K. Tehran, (2014) Process optimization for fluoride remove from water by moringaoleifera seed extract ***Fluoride***, 47: 152-160.
- [9]. M. Ghaderpoori, N. Ali Asghar, G. Afsin, S. Mahmoud, (2018) Heavy metal analysis and quality assessment in drinking water Khorramabad city, Iran, ***Data in Brief***, 16: 685-92,
- [10]. V. Sunitha, B. Rajeswara Reddy, M.R.K. Reddy, (2012) Groundwater Quality Evaluation with special Reference to fluoride and nitrate Pollution in Uravakonda, Anantapur District, Andhra Pradesh- a case study Andhra Pradesh, ***India International Journal of Research in Chemistry and Environment***, 2(1), 88-96.
- [11]. B. Muralidhara Reddy, V. Sunitha, M.R.K. Reddy (2013) Fluoride and Nitrate Geochemistry of Groundwater from Kadiri, Mudigubba and Nallamada Mandals of Anantapur District, Andhra Pradesh, India, ***Journal of Agricultural Engineering and Biotechnology***, 1(2):37-42.
- [12]. B. Muralidhara Reddy, V. Sunitha, M. Prasad, Y. Sudharshan Reddy, M.R.K. Reddy, (2019) Evaluation of groundwater suitability for domestic and agricultural utility in arid region of Anantapur, Andhra Pradesh State, South India. ***Groundwater for Sustainable Development***, 9: 100262.
- [13]. V. Sunitha, B. Muralidhara Reddy, (2014) Determination of Fluoride Concentration in Ground Water by Ion Selective Electrode, ***International Journal of Current Research and Academic Review***, 2 (8): 159-166.

- [14]. V. Sunitha, J. Abdullah Khan, B. Muralidhara Reddy, M. Prasad, M.R.K. Reddy, (2014) Assessment of Groundwater Quality in Parts of Kadapa and Anantapur Districts, Andhra Pradesh, India, **Indian Journal of Advances in Chemical Science**, 3: 96-101.
- [15]. V. Sunitha, B. Muralidhara Reddy, S. Sumithra (2016). Assessment of Groundwater Quality Index in the Kadapa Municipal City, Y.S.R. District, Andhra Pradesh **International Journal of Advanced Research**, 6 (2): 545-548.
- [16]. V. Sunitha, B. MuralidharaReddy, (2018) Defluoridation of water using mentha longifolia (Mint) as Bioadsorbent, **Journal of Geophysical Union**, 22(2):207-211.
- [17]. M. Prasad, V. Sunitha, Y. Sudharshan Reddy, B. Suvarna, B. Muralidhara Reddy, M.R.K. Reddy, (2019) Data on water quality index development for groundwater quality assessment from Obulavaripalli Mandal, YSR district, AP India, **Data in brief**, 24: 103846.
- [18]. Y. Sudharshan Reddy, V. Sunitha, B. Suvarna, (2020) Groundwater quality evaluation using GIS and water quality index in and around inactive mines, Southwestern parts of Cuddapah basin, Andhra Pradesh, South India, **Hydro Research**, 3:146-157.
- [19]. Y. Sudharshan Reddy, V. Sunitha, B. Suvarna, (2020) Monitoring of Groundwater quality for drinking purposes using the WQI method and its health implications around inactive mines in Vemula - Vemaplli region, Kadapa District, South India, **Applied Water Science**, 10:202.
- [20]. Y. Sudharshan Reddy, V. Sunitha, B. Suvarna, M. Prasad, M. Muralidhara Reddy, M.R.K. Reddy, (2020) Data on Physical impacts and hydrogeochemical assessment of Inactive/Abandoned mines in and around Southwestern parts of the Cuddapah basin using a conceptual site model (CSM) **Data in Brief**, 29: 105187.
- [21]. Y. Sudharshan Reddy, V. Sunitha (2020) Development of water quality index for groundwater quality evaluation in and around Vemula Mandal, YSR District (A.P., India), **Journal of Indian Geophysical Union**, 24(1): 1-29.
- [22]. Y. Sudharshan Reddy, V. Sunitha, (2020) Application of relative weight method for groundwater quality assessment in Northeastern part of Y.S.R District, A.P India, **Indian Journal of advances in Chemical Sciences**, 8:22-27.
- [23]. B. Suvarna, Y. Sudharsan Reddy, V. Sunitha, B. MuralidharaReddy, M. Prasa, M.R.K. Reddy, (2019) Data on application of water quality index method for appraisal of water quality in around cement industrial corridor, Yerraguntla Mandal, Y.S.R District,A.P South India, **Data in Brief**, 28: 104872.
- [24]. B. Suvarna, V. Sunitha, Y. Sudharshan Reddy, M.R.K. Reddy, (2020) Data on health risk assessment of nitrate contamination in groundwater of rural region in the Yerraguntla Mandal, South India, **Data in Brief**, 30: 105374.
- [25]. B. Muralidhar Reddy, V. Sunitha, (2020) Geochemical and health risk assessment of fluoride and nitrate toxicity in semi- arid region of Anantapur District, South India. **Environmental Chemistry and Ecotoxicology**, 2: 150-161.
- [26]. N. Subba Rao (2011) High-fluoride groundwater. **Environmental Monitoring and Assessment**, 176 (1-4):637-645.

- [27]. N. Subba Rao N, (2017) Controlling factors of fluoride in groundwater in a part of south India. *Arabian Journal of Geosciences*, 10 (23):1-15.
- [28]. J.D. Hem (1991) Study and interpretation of the chemical characteristics of natural water. 3rd edition, *Scientific Publishers*, 2254.
- [29]. S.K. Swain, S. Mishra, T. Patnaik, R.K. Patel, Jha Usha, R.K. Dey (2012) Fluoride removal performance of a new hybrid sorbent of Zr(IV)-ethylenediamine, *Chemical Engineering Journal*, 184: 72-81.
- [30]. W.X. Gong, J.H. Qua, R.P. Liu, H.C. Lan (2012) Effect of aluminum fluoride complexation on fluoride removal by coagulation, *Colloids and Surfaces A: Physicochemical and Engineering Aspects*, 395: 88-93.
- [31]. M.G. Sujana. S. Anand (2011) Fluoride removal studies from contaminated ground water by using bauxite, *Desalination*, 267: 222-227.
- [32]. S. Kagne, S. Jagtap, P. Dhawade, S.P. Kamble, S. Devotta, S.S. Rayalu, (2008) Hydrated cement: a promising adsorbent for the removal of fluoride from aqueous solution, *J Journal of hazardous materials*, 154: 88-95.
- [33]. X. Wu, Y. Zhang, X. Dou, M. Yang (2009) Fluoride removal performance of a novel Fe–Al–Cetrimetal oxide adsorbent, *Chemosphere*, 69: 1758–1764.
- [34]. K. Biswas, K. Gupta, A. Goswami, U.C. Ghosh, (2010) Fluoride removal efficiency from aqueous solution by synthetic iron (III)- aluminum (III)-Chromium (III) ternary mixed oxide, *Desalination*, 255: 44–51.
- [35]. N. Viswanathan, S. Meenakshi, (2010) Selective fluoride adsorption by a hydrotalcite/chitosan composite, *Applied Clay Science*, 48: 607–611.
- [36]. M.S. Frant, J.W. Ross, (1966) Electrode for sensing fluoride ion activity in solution. *Science*, 154, 1553-1555.
- [37]. M.S. Frant, (1994) History of the Early Commercialization of Ion-Selective Electrodes, *Analyst*, 199: 2293-2301.
- [38]. A. Kumar, A. Balouch, Abdullah. (2021) Remediation of toxic fluoride from aqueous media by various techniques. *International Journal of Environmental Analytical Chemistry*, 101(4):482-505.
- [39]. A. Dhillon, S. Prasad, D. Kumar (2017) Recent advances and spectroscopic perspectives in fluoride removal. *Applied Spectroscopy Reviews* 52(3):175-230
- [40]. V.K. Rao, (1997) Defluoridation of drinking water by Prasanthi technique. *National Workshop on defluoridation technologies for Fluorosis control*. 28.
- [41]. K. Hu, J.M. Dickson (2006) Nanofiltration membrane performance on fluoride removal from water, *Journal of Membrane Science*, 279: 529–538.
- [42]. S. Bason, A. Ben-David, Y. Oren, V. Freger, (2006). Characterization of ion transport in the active layer of RO and NF polyamide membranes, *Desalination*, 199: 31–33.
- [43]. A. Szymczyk, P. Fievet, (2005). Investigating transport properties of nanofiltration membranes by means of a steric, electric and dielectric exclusion model. *Journal of Membrane Science*, 252: 77–88.

- [44]. X. Lefebvre, J. Palmeri, (2005) Nanofiltration theory: good co-ion exclusion approximation for single salts, *Journal of Physical Chemistry B*, 109: 5525–5540.
- [45]. J. Cervera, V. Garcia-Morales, J. Pellicer, (2003) Ion size effects on the electrokinetic flow in nanoporous membranes caused by concentration gradients, *Journal of Physical Chemistry B* 107: 8300–8309
- [46]. J. Palmeri, P. Blanc, A. Larbot, P. David, (1999) Theory of pressure- driven transport of neutral solutes and ions in porous ceramic nanofiltration membranes. *Journal of Membrane Science*, 160: 141–170.
- [47]. S.S. Tripathy, A.M. Raichur, (2008) Abatement of fluoride from water using manganese dioxide-coated activated alumina. *Journal of Hazardous Materials*, 153(3):1043-1051.
- [48]. A.K. Shakya, R. Bhande, P.K. Ghosh, (2020) A practical approach on reuse of drinking water treatment plant residuals for fluoride removal. *Environmental technology*, 41(22): 2907-2919.
- [49]. S. Kumari, S. Khan, (2017) Defluoridation technology for drinking water and tea by green synthesized $\text{Fe}_3\text{O}_4/\text{Al}_2\text{O}_3$ nanoparticles coated polyurethane foams for rural communities. *Scientific reports*, 7(1): 1-12.
- [50]. M.R. Khairnar, A.S. Dodamani, H.C. Jadhav, R.G. Naik, M.A. Deshmukh, (2015) Mitigation of fluorosis-a review. *Journal of clinical and diagnostic research*, 9(6): ZE05.
- [51]. A.K. Susheela (1992) Defluoridation of drinking water—merits of alternative technologies. *A Nutrition Foundation of India*.
- [52]. N.S. Gadhari, B.J. Sanghavi, S.P. Karna, A.K. Srivastava, 2010. Potentiometric stripping analysis of bismuth based on carbon paste electrode modified with cryptand [2.2.1] and multiwalled carbon nanotubes. *Electrochimica Acta*, 56(2): 627-635.
- [53]. B.J. Sanghavi, A.K. Srivastava, (2011) Adsorptive stripping differential pulse voltammetric determination of venlafaxine and desvenlafaxine employing Nafion–carbon nanotube composite glassy carbon electrode. *Electrochimica Acta*, 56(11): 4188-4196.
- [54]. S.M. Mobin, B.J. Sanghavi, A.K. Srivastava, P. Mathur, G.K. Lahiri, (2010) Biomimetic sensor for certain phenols employing a copper (II) complex. *Analytical chemistry*, 82(14): 5983-5992.
- [55]. R.M. Kotkar, P.B. Desai, A.K. Srivastava, (2007) Behavior of riboflavin on plain carbon paste and aza macrocycles based chemically modified electrodes. *Sensors and Actuators B: Chemical*, 124(1): 90-98.
- [56]. S. Shahrokhian, M. Ghalkhani, M.K. Amini, (2009) Application of carbon-paste electrode modified with iron phthalocyanine for voltammetric determination of epinephrine in the presence of ascorbic acid and uric acid. *Sensors and Actuators B: Chemical*, 137(2): 669-675.
- [57]. Y. Ma, S.G. Wang, M. Fan, W.X. Gong, B.Y. Gao, (2009) Characteristics and defluoridation performance of granular activated carbons coated with manganese oxides. *Journal of hazardous materials*, 168(2-3): 1140-1146.

- [58]. S. Shahrokhian, M. Ghalkhani, M.K. Amini, (2009) Application of carbon-paste electrode modified with iron phthalocyanine for voltammetric determination of epinephrine in the presence of ascorbic acid and uric acid. ***Sensors and Actuators B: Chemical***, 137(2): 669-675.
- [59]. D. Mohapatra, D. Mishra, S.P. Mishra, G.R. Chaudhury, R.P. Das (2004) Use of oxide minerals to abate fluoride from water. ***Journal of colloid and interface science***, 275(2): 355-359.
- [60]. L. Zhang, Z. Shi, Q. Lang, J. Pan, (2010) Electrochemical synthesis of belt-like polyaniline network on p-phenylenediamine functionalized glassy carbon electrode and its use for the direct electrochemistry of horse heart cytochrome C. ***Electrochimica acta***, 55(3): 641-647.
- [61]. I.R.W. de Oliveira, R.E.H.M. de Barros, A. Neves, I.C. Vieira, (2007) Biomimetic sensor based on a novel copper complex for the determination of hydroquinone in cosmetics. ***Sensors and Actuators B: Chemical***, 122(1): 89-94.
- [62]. T. Yang, N. Zhou, Y. Zhang, W. Zhang, K. Jiao, G. Li, (2009) Synergistically improved sensitivity for the detection of specific DNA sequences using polyaniline nanofibers and multi-walled carbon nanotubes composites. ***Biosensors and Bioelectronics***, 24(7): 2165-2170.
- [63]. M.G. Sujana, S. Anand (2011) Fluoride removal studies from contaminated ground water by using bauxite, ***Desalination***, 267: 222–227.
- [64]. S.I. Alhassan, Y. He, L. Huang, B. Wu, L. Yan, H. Deng, H. Wang, (2020) A review on fluoride adsorption using modified bauxite: Surface modification and sorption mechanisms perspectives. ***Journal of Environmental Chemical Engineering*** 8: 104532
- [65]. R. Lavecchia, F. Medici, L. Piga, G. Rinaldi, A. Zuurro (2012) Fluoride removal from water by adsorption on a high alumina content bauxite. ***Chemical Engineering***; 26:225-230.
- [66]. A.V. Jamode, V.S. Sapkal, V.S. Jamode, (2004) Defluoridation of water using inexpensive adsorbents, ***Journal of the Indian Institute of Science***, 84: 163-171.
- [67]. Y. Wang, J. Reardon Eric (2001) Activation and regeneration of a soil sorbent for defluoridation of drinking water, ***Applied Geochemistry***, 16: 531–539.
- [68]. Beitollahi H, Karimi-Maleh H, Khabazzadeh H. (2008) Nanomolar and selective determination of epinephrine in the presence of norepinephrine using carbon paste electrode modified with carbon nanotubes and novel 2-(4-oxo-3-phenyl-3, 4-dihydroquinazolinyl)-N'-phenyl-hydrazinecarbothioamide. ***Analytical Chemistry***. 80(24):9848-9851.